



Report

Kriti '26

Drone Design Challenge

Hostel ID:

2601

MARUT
- DRONES -

1 Introduction

This document provides comprehensive mechanical design documentation for the Seedball Dispenser Module developed as part of the Kriti26 project. The system is a drone-mounted payload designed to autonomously deploy seedballs for aerial reforestation and land restoration applications.

The mechanical assembly comprises five primary sub-components — the Storage Module, the Hopper (with the Hopper with Servo Mounts being its detailed production-ready iteration), the V-File Channel, and the Connection Bracket — all designed in Autodesk Fusion 360 and manufactured to meet the dimensional tolerances specified within the engineering drawings.

1.1 Purpose of This Document

This design documentation serves as the primary reference for the mechanical sub-system, intended for use by the manufacturing team, integration engineers, and future maintainers. It covers:

- Functional description of each sub-component
- Dimensional specifications and tolerances extracted from engineering drawings
- Bounding volume and mesh complexity data from CAD files
- Assembly relationships and interface definitions
- File format inventory and CAD file structure
- Design rationale, constraints, and assumptions
- Manufacturing and assembly guidelines

1.2 Scope

This document covers only the mechanical design of the dispensing module. Electronics integration (ESP32 firmware, servo wiring, IR sensor placement), ROS 2 simulation, and flight controller integration are documented separately in their respective sub-system documents.

1.3 Document Conventions

All linear dimensions in this document are expressed in millimetres (mm) unless otherwise stated. Angular dimensions are expressed in degrees (°). Tolerances are expressed in the $\pm$ format as defined in the engineering drawings. File paths use forward slashes and are relative to the project root.

2 System Architecture & Design Intent

2.1 Functional Overview

The Seedball Dispenser Module is a gravity-assisted dispensing mechanism. Seedballs are pre-loaded into the storage hopper prior to flight. During operation, the ESP32 microcontroller receives a drop command via UART from the onboard computer (Jetson/Raspberry Pi) and actuates one of two servo-controlled gates at the base of the hopper. Gravity causes seedballs to exit the hopper through a V-shaped channel that guides them to the drop point. IR sensors along the channel and at the neck confirm successful dispensing and detect jams.

The dual-file design (File A / File B, corresponding to Servo A and Servo B) allows the system to switch to a secondary channel if the primary becomes jammed, increasing mission reliability without requiring a physical

intervention.

2.2 Design Principles

- **Lightweight construction:** Core structural components are fabricated from Aluminium Alloy (4xx.x Series), providing an optimal strength-to-weight ratio. The total calculated mass of the hopper assembly is restricted to an optimised 2.85 kg.
- **Modularity:** Each sub-component is independently replaceable. Standardised mating flanges and fastener interfaces allow sub-components to be re-cast or re-formed individually without disassembling the full stack.
- **Gravity-fed operation:** No additional actuators are required for ball transport; only the gate servos consume power.
- **Dual-channel redundancy:** Two independent dispensing files extend operational continuity during jams.
- **Tool-free assembly:** Connection brackets and mounting interfaces are designed for snap-fit or screw-based assembly accessible without specialised tooling.

2.3 Mechanical Sub-System Hierarchy

The full assembly is composed of five interacting sub-assemblies arranged in a vertical stack:

#	Sub-Component	Role in Assembly	CAD File
1	Storage Module	Top-level bulk storage reservoir for seedballs	03_STORAGE.f3d
2	Hopper	Collects and funnels balls from storage to gate	01_HOPPER.f3d
2a	Hopper with Servo Mounts	Production-ready iteration of the Hopper — same base geometry with integrated servo mounting detail	02_HOPPER_WITH-SERVO-MOUNTS.f3d
4	V-File Channel	Dual V-shaped guide channel directing balls to exit	04_V-FILE.f3d
5	Connection Bracket	Structural interface connecting module to drone frame	05_CONNECTION.f3d

Table 1: CAD Sub-Components

3 Component Design Specifications

3.1 Storage Module

The Storage Module is the topmost component in the assembly stack. It serves as the primary seedball reservoir, holding the full payload of balls prior to dispensing. The module interfaces with the Hopper below it via a mating flange, allowing balls to feed through under gravity.

3.1.1 Dimensional Specifications

Parameter	Value	Tolerance
Overall Width (X)	294.7 mm	±0.5 mm
Overall Depth (Y)	317.5 mm	±0.5 mm
Overall Height (Z)	105.0 mm	±0.3 mm
Front Face Width	295 mm	±0.5 mm
Side Panel Depth	317.5 mm	±0.5 mm
Bottom Opening Width	160 mm	±0.4 mm
Mounting Hole Diameter	Ø6.5 mm	—
Hole Offset from Edge	17.5 mm	—

The storage module features a rectangular cross-section with a tapered internal floor to direct balls toward the central outlet. The Ø6.5 mm mounting holes positioned 17.5 mm from the edge provide the structural connection to the hopper body below.

3.2 Hopper / Hopper with Servo Mounts

The Hopper and the Hopper with Servo Mounts share the same base CAD geometry. The Hopper (01_HOPPER) represents the foundational design, while the Hopper with Servo Mounts (02_HOPPER_WITH-SERVO-MOUNTS) is the fully detailed, production-ready iteration — it adds integrated servo mounting bosses to the base body without altering the core hopper geometry. The Hopper with Servo Mounts is the version used in the final assembly.

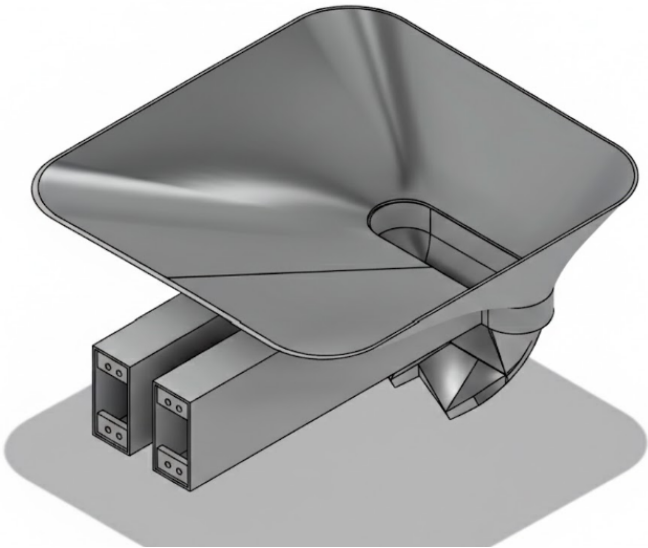


Figure 1: Hopper with servo mount

The Hopper receives seedballs from the Storage Module and funnels them into the dual-file channel. Its angled internal walls are designed to prevent bridging — a common failure mode in granular dispensers where balls interlock and block the outlet. The geometry transitions from a wide rectangular input at the top to a narrower rectangular exit at the base.

3.2.1 Dimensional Specifications — Base Hopper (01_HOPPER)

Parameter	Value	Tolerance
Overall Width (X)	294.7 mm	±0.5 mm
Overall Depth (Y)	317.5 mm	±0.5 mm
Overall Height (Z)	133.1 mm	—
Top Opening Width	295 mm	±0.5 mm
Base Width	317 mm	±0.5 mm
Wall Taper Angle	53.0° ±0.7° / 57.5° ±1°	±1°
Bottom Flat Width	50 mm	—
Side Reference Dim	150 mm	±0.4 mm
Internal Radius (base corner)	R23 mm	—

The dual-angle internal walls (53° and 57.5°) were selected based on the expected diameter range of seedballs (15–30 mm) to ensure reliable flow without compaction. The R23 mm internal radius at the hopper base reduces stress concentrations and prevents ball jamming at the outlet corner.

3.2.2 Hopper with Servo Mounts — Additional Dimensions

The following dimensions apply specifically to the Hopper with Servo Mounts (02_HOPPER_WITH-SERVO-MOUNTS), reflecting the additional servo mounting geometry added to the base hopper body.

3.2.3 Servo Mount Interface Notes

Parameter	Value	Tolerance
Overall Width (X)	294.7 mm	±0.5 mm
Overall Depth (Y)	335.4 mm	±0.5 mm
Overall Height (Z)	213.2 mm	—
Mount Plate Width	156 mm	±0.4 mm
Upper Body Width	263.6 mm	±0.5 mm
Servo Hole Diameter	Ø4.5 mm	—
Mount Hole Spacing A	94 mm	—
Mount Hole Spacing B	96 mm	—
Bracket Tab Height	34 mm	—
Bracket Tab Depth	10 mm	—
Mount Offset from Top	21 mm	—

The Ø4.5 mm servo mounting holes are sized for M4 bolts or self-tapping screws commonly used with standard SG90/MG90S class servos. The 94 mm and 96 mm hole spacings correspond to the standard servo body dimensions, ensuring compatibility with off-the-shelf servo hardware without modification.

3.3 V-File Channel

The V-File Channel is the core dispensing element. It is a dual-channel guide that sits immediately below the hopper gate exit. Each V-shaped trough corresponds to one servo gate (File A or File B). Seedballs roll along the angled walls of the V-channel under gravity and exit at the bottom opening when the gate is open. IR sensors are positioned at the start, end, and neck of each channel to monitor ball flow and detect jam conditions.

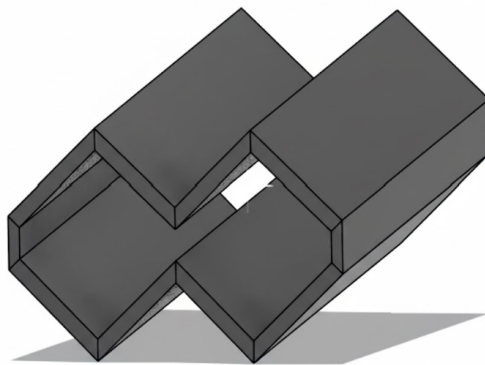


Figure 2: V File

3.3.1 Dimensional Specifications

Parameter	Value	Tolerance
Overall Width (X)	107.1 mm	± 0.33 mm
Overall Depth (Y)	256.4 mm	± 0.5 mm
Overall Height (Z)	95.4 mm	—
Channel Length	251 mm	± 0.5 mm
Channel Width	37 mm	± 0.4 mm
V-Angle	90°	—
Wall Height	71 mm	—
Channel Base Radius	R5.152 mm	—
Outer Profile Radius	R6.37 mm	—
Partition Wall Thickness	107 mm	± 0.33 mm

The 90° V-angle was chosen to be compatible with seedballs ranging from 15–30 mm in diameter, providing stable seating and controlled rolling motion. The R5.152 mm base radius prevents smaller balls from compacting at the channel bottom. The 251 mm channel length provides adequate travel distance for IR sensor placement with sufficient separation between the start and end sensor positions.

3.4 Connection Bracket

The Connection Bracket is the structural interface that attaches the dispenser module to the drone's payload mount or underbelly frame. It is designed to accommodate standard drone mounting rail configurations and provides the primary mechanical load path during flight. The bracket also acts as the bottom closure of the

V-File assembly.

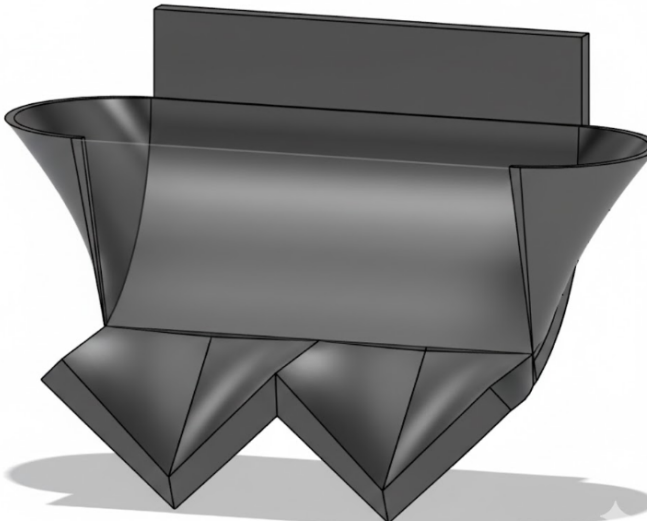


Figure 3: Connection Bracket

3.4.1 Dimensional Specifications

Parameter	Value	Tolerance
Overall Width (X)	150.0 mm	±0.4 mm
Overall Depth (Y)	68.6 mm	—
Overall Height (Z)	100.0 mm	—
Bracket Width	150 mm	±0.4 mm
Tab Height	54 mm	±0.2 mm
Outer Radius (top)	R50 mm	—
Inner Radius A	R21 mm	—
Inner Radius B	R23 mm	—
Lower Corner Radius	R25 mm	—

The R50 mm outer profile at the top of the bracket provides a smooth curved interface with the drone frame, reducing stress concentrations under vibration loads during flight. The multiple internal radii (R21, R23, R25) are filleted transitions that maintain structural integrity while minimising material usage.

4 Assembly & Interface Definitions

4.1 Assembly Stack Order

The module is assembled in the following order from top to bottom. Each component interfaces with the one below it via mating flanges, alignment features, or mechanical fasteners as described:

Order	Component	Interface Type	Fastener / Feature
1 (Top)	Storage Module	Flange-to-flange with Hopper	M3/M4 screws through Ø6.5 mm holes
2	Hopper with Servo Mounts	Receives storage module above, V-file below	Servo mount bosses (Ø4.5 mm), M4 bolts
3	V-File Channel	Slides into hopper base exit	Press-fit or adhesive bond
4 (Bottom)	Connection Bracket	Interfaces with drone mount rails	M4/M5 bolts through bracket tabs

4.2 Mating Dimensions & Clearances

The Storage Module base opening (160 mm wide) aligns with the Hopper's top inlet. The 17.5 mm hole offset on the Storage Module corresponds to matching bosses on the Hopper top flange. All mating interfaces are manufactured to the die-casting tolerance standard $T = 0.25 + 0.025(L - 1)$, where L is the dimension in inches, ensuring consistent fit across production batches.

The V-File Channel exit width (107.1 mm) is sized to fit within the 156 mm mount plate width of the Hopper with Servo Mounts, leaving clearance on both sides for the servo arms to travel their full angular range without obstruction.

4.3 Servo Integration Points

The two servo gate axes are located at the base of the Hopper with Servo Mounts component. Servo A controls File A (left channel) and Servo B controls File B (right channel) as referenced in the ESP32 firmware. The Ø4.5 mm mounting holes with 94 mm and 96 mm spacings accept standard micro-servo flanges. Servo horns extend horizontally to actuate the gate mechanism inside the V-File Channel.

4.4 Sensor Mounting Positions

Five IR sensors are positioned along the assembly as follows. All sensors interface with the ESP32 via the IR Sensor Array power rail (3.5 V) and signal lines routed along the module's exterior walls. Sensor positions should be confirmed against the actual part before final wiring:

1. IR_A_START (GPIO 36): Positioned at the inlet of File A channel to detect ball entry.
2. IR_A_END (GPIO 39): Positioned at the outlet of File A channel to confirm ball exit.
3. IR_B_START (GPIO 34): Positioned at the inlet of File B channel to detect ball entry.
4. IR_B_END (GPIO 35): Positioned at the outlet of File B channel to confirm ball exit.
5. IR_NECK (GPIO 32): Positioned at the neck/exit point common to both files to confirm final dispensing.

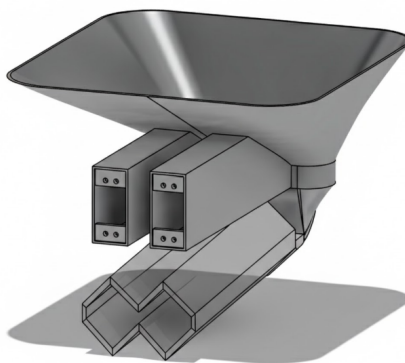


Figure 4: Assembly

5 CAD File Inventory

5.1 Directory Structure

The CAD and drawing files are organised into two directories as follows:

CAD_AND_DRAWING/

01_CAD_FILES/ -- Fusion 360 native, STEP, and STL files

01_DRAWING/ -- Engineering drawings (PDF and DXF)

5.2 CAD Files

Filename	Format	Description	Tris
00_FULL-ASSEMBLY.f3z	Fusion Archive	Complete assembly with all sub-components	—
01_HOPPER .f3d/.step/.stl	Multi-format	Base hopper body only	20,668
02_HOPPER_WITH-SERVO-MOUNTS.f3d/.step/.stl	Multi-format	Hopper with integrated servo mounts	30,134
03_STORAGE.f3d/.step/.stl	Multi-format	Storage reservoir module	2,832
04_V-FILE .f3d/.step/.stl	Multi-format	Dual V-channel dispenser	112
05_CONNECTION.f3d/.step/.stl	Multi-format	Drone mounting bracket	11,948

5.3 Engineering Drawing Files

Drawing File	PDF	DXF	Component
01_HOPPER	✓	✓	Hopper
02_HOPPER_WITH-MOUNTS	✓	✓	Hopper with Servo Mounts
03_STORAGE	✓	✓	Storage Module
04_V-FILE	✓	✓	V-File Channel
05_CONNECTION	✓	✓	Connection Bracket

5.4 File Format Notes

Each sub-component is provided in three export formats to support different downstream workflows. The **.f3d** (Fusion 360 native) files retain full parametric history and sketch constraints and should be used for any design modifications. The **.step** (AP214) files are format-agnostic and can be imported into any major CAD tool (SolidWorks, FreeCAD, Onshape, CATIA). The **.stl** files are triangulated surface meshes exported at fine resolution for direct use in slicers or simulation tools such as Gazebo.

The full assembly archive **.f3z** includes all sub-components with their assembly constraints and is the single source of truth for the complete mechanical design.

6 Material Selection & Fabrication Methodology

6.1 Material Selection Rationale

The hopper assembly is dynamically mounted to a UAV frame and functions as an active payload deployment system. The engineering constraints mandate a strict balance between mass reduction and structural integrity to ensure both flight stability and operational reliability over rugged terrain.

6.1.1 Materials Evaluated and Rejected

Two material classes were evaluated and rejected prior to the final selection:

- **Additive Manufacturing Polymers (PLA, ABS, PETG):** While polymers offer significant mass reduction, they inherently lack the ultimate tensile strength and stiffness required for a structural reservoir under flight loads. Standard FDM/SLA polymers also possess poor vibration attenuation characteristics. The harmonic vibrations generated by UAV rotors would propagate through the polymer chassis and potentially disrupt seedball singulation in the V-File channel.
- **High-Strength Ferrous Metals (Steel):** Steel provides superior dimensional stability and structural strength. However, its high density incurs an unacceptable weight penalty that would severely degrade the UAV’s manoeuvrability, battery life, and total seedball payload capacity.

6.1.2 Final Selection: Aluminium Alloy (4xx.x Series)

Aluminium Alloy of the 4xx.x series was selected as the definitive material for all core structural components, bridging the gap between the two primary mechanical requirements. The 4xx.x series was specifically chosen for its superior fluidity and minimal solidification shrinkage, making it the industry standard for complex casting geometries.

Property	Value / Description
Material	Aluminium Alloy — 4xx.x Series
Total Assembly Mass	2.85 kg (inclusive of storage unit and transition section)
Density Advantage	Low material density; optimal strength-to-weight ratio
Corrosion Resistance	Excellent — natural passivation layer; critical for soil, moisture, and clay exposure
Machinability	High machinability and weldability for secondary operations

6.2 Fabrication Methodology

The fabrication strategy is divided into two distinct processes based on the geometric complexity, cross-sectional variance, and structural requirements of the individual sub-components.

6.2.1 Primary Assembly Fabrication — Cold Chamber Die Casting

The core hopper architecture, upper storage unit, servo motor mounts, and intermediate transition parts feature complex curved surfaces, specific draft angles, and varying wall thicknesses. These geometries render standard subtractive machining (CNC) inefficient. Cold Chamber Die Casting was identified as the most appropriate manufacturing process, outperforming sand or investment casting by producing a superior surface finish that minimises internal wall friction and causes fewer thermal distortions.

- **Cold Chamber Process:** Because aluminium alloys require high processing temperatures that would degrade a standard hot-chamber pumping system, the cold chamber variant is used throughout.
- **Execution:** The 4xx.x aluminium alloy is melted in an external furnace. The molten metal is ladled into an injection chamber, and a hydraulic piston forces the liquid metal into the two halves of a hardened steel mould cavity under extremely high pressure, ensuring complete filling of the asymmetric hopper walls before solidification.
- **Applicable components:** Hopper body, Hopper with Servo Mounts, Storage Module, Connection Bracket.

6.2.2 Secondary Component Fabrication — Sheet Metal Forming

The V-File component (the singulation channel) features a strictly planar, less geometrically complex architecture; casting would therefore be an unnecessary expense.

- **Forming:** The V-File is manufactured via standard sheet metal bending operations using aluminium stock with a uniform thickness of 2 mm.
- **Integration:** Post-forming, the V-File channel is permanently affixed to the intermediate die-cast section of the hopper via structural welding, ensuring a rigid, seamless transition for seedballs from the hopper outlet into the metering mechanism.

6.3 Dimensional Tolerances

To ensure precise mechanical assembly, reliable kinematic function of the servo escapement, and repeatable mass flow, strict dimensional controls must be maintained during fabrication. Standard industrial die-casting tolerances have been applied to account for thermal expansion and tool wear.

6.3.1 Tolerance Formula

For any given feature dimension expressed in inches (L), the allowable manufacturing tolerance in mm is calculated using the standard industrial die-casting equation:

$$T = 0.25 + 0.025 \times (L - 1)$$

Where T is the permissible tolerance (mm) and L is the feature dimension (inches). The base tolerance for the first inch is ± 0.25 mm, with an incremental ± 0.025 mm applied for every additional inch of linear length.

6.3.2 Tolerance Reference Table

Tolerance Class	Value	Applied To
Standard linear	± 0.5 mm	Overall body dimensions (width, depth, height)
Fit-critical linear	± 0.3 – 0.4 mm	Mating faces, outlet widths, sensor mount zones
Angular	$\pm 1^\circ$	Hopper funnel wall angles (53° , 57.5°)
Tight angular	$\pm 0.7^\circ$	Primary funnel angle (53.0°)
Hole diameter	Per drawing	Servo holes $\varnothing 4.5$ mm, mounting holes $\varnothing 6.5$ mm
Die-casting base	± 0.25 mm / first inch	All die-cast features
Die-casting incremental	$+0.025$ mm / additional inch	All die-cast features beyond 1 inch

These calculated tolerances have been comprehensively incorporated into the technical 2D CAD engineering drawings, dictating the acceptable limits for tooling and post-casting quality control inspections.

6.4 Post-Fabrication & Assembly Guidelines

1. Inspect all die-cast surfaces for porosity, cold shuts, or misruns before proceeding to assembly. Reject any part with visible surface defects on internal funnel walls.
2. Machine the servo mounting holes ($\varnothing 4.5$ mm) to final tolerance post-casting, as cast holes may require a light reaming pass to achieve the required fit.
3. Verify the V-File 90° V-angle with a precision square tool after sheet metal bending. Angular deviation directly affects seedball guidance and singulation accuracy.
4. Clean the V-File channel interior thoroughly before welding to the hopper to prevent weld contamination

from organic debris.

5. Perform a post-weld visual and dimensional inspection on the V-File to hopper joint to confirm alignment of the channel centre-line with the hopper outlet axis.

7 Design Rationale, Constraints & Future Work

7.1 Key Design Decisions

Several design decisions warrant specific documentation to support future modifications:

7.1.1 Dual-File Channel Architecture

The decision to implement two independent dispensing channels (File A and File B) rather than a single channel was driven by the need for mission continuity. A single-channel design would require the drone to return to base if a jam occurred. The dual-file architecture allows the firmware to switch to the secondary file mid-mission, effectively doubling the available mean-time-before-failure for the mechanical dispenser under field conditions.

7.1.2 Gravity-Fed Operation

A purely gravity-fed design was chosen over motorised conveyor or auger alternatives to minimise electrical power consumption and mechanical complexity. Given the seedball diameter range (15–30 mm) and the module's operating altitude (low-speed hover), gravity provides sufficient and controllable dispensing force through the funnel geometry. The angled hopper walls and V-channel work together to regulate flow rate passively.

7.1.3 Modular Stack Design

Each sub-component is independently replaceable without requiring full disassembly of the module. The standardised 295 mm × 317.5 mm footprint shared between the Storage Module and the Hopper ensures cross-component compatibility. The Hopper with Servo Mounts is the production version of the Hopper and is fully interchangeable with it at the assembly level.

7.2 Known Constraints & Limitations

- The current design is sized for seedballs of 15–30 mm diameter. Balls outside this range may cause bridging at the hopper base or insufficient guidance in the V-channel.
- Die-cast aluminium parts may exhibit minor surface porosity in non-critical zones; however, internal funnel walls and servo mating surfaces must be free of porosity defects that could affect ball flow or fastener retention.
- The V-File Channel has a relatively low STL triangle count (112 triangles), indicating a simplified geometry. This may require refinement if tighter ball guidance or sensor mounting bosses are needed.
- The Storage Module height (105 mm) limits the total seedball payload. Increasing this dimension requires a corresponding check on drone payload capacity and centre-of-gravity balance.

7.3 Future Work & Design Improvements

Item	Description	Priority
Sensor mounting bosses	Add integrated snap-fit bosses for IR sensor placement directly to V-File and Hopper parts	High
Increased storage height	Evaluate extending Storage Module height to 150+ mm for larger payload	Medium
Weight analysis	Perform mass properties analysis in Fusion 360 and validate against drone payload spec	High
Wire routing channels	Add recessed channels on component exteriors for clean servo and sensor wire routing	Medium
Ball-size adaptability	Parametric CAD model with ball-diameter input to auto-adjust funnel angles	Low
Vibration testing	Bench-test full die-cast assembly on vibration rig to validate weld joints and fastener retention at UAV rotor frequencies	High
Gate seal design	Investigate adding a soft gasket to servo gate to prevent ball roll-out when gate is nominally closed	Medium